

Our transcriptional studies for aneuploids bearing duplications for the regions 16F-20, 13F-18A and 8C-20, of the X chromosome show that at the transcriptional level monitored by autoradiography (Lakhotia & Mukherjee 1969), the activity of the X chromosome (relative to autosome) segments is significantly reduced when the concerned segment is not included within the duplicated region (Table 1).

The activity of two X-coded enzymes, [G6PD (18DE) and 6PGD (2D3)] normalized against an autosomal enzyme - [NADP dependent IDH (66A-D)] was studied for the aneuploids for (18A-20), (16F-20), (15EF-20), and (13F-18A) regions of the X chromosome, according to the method of Lucchesi & Rawls (1973). The normalized 6PGD activity of those aneuploids duplicated for the proximal region of the X chromosome was not decreased when compared to control levels of Oregon R⁺ male and female. Rather, a significant increase in the value was noted for Dp(16F-20) male, Dp(13F-18A) male and for Dp(13F-18A) female, even though they had the normal dose of 6PGD gene (Table 2). Thus, dosage compensation might involve the processing of X-coded transcripts at their translational level to vary the apparent number of enzyme molecules.

References. Lakhotia, S.C. & A.S. Mukherjee 1969, Genet. Res. 14:137-150; Lucchesi, J.C. & R.H. Rawls 1973, Biochem. Genet. 91:41-51; Maroni, G. & J.C. Lucchesi 1980, Chromosoma (Berl) 77:253-261; Stewart, B.R. & J.R. Merriam 1975, Genetics 79:635-647.

Gilbert, D.G. P.O. Box 302, Bloomington, Indiana. Estimating single gene effects on quantitative traits. 3. Test of the single population method with *D.melanogaster*.

A wild type population is examined for effects of 5 allozymes on 6 quantitative traits using the single population method (Gilbert 1985). Single or multiple locus genetic variation for each trait is demonstrated.

The population of *D.melanogaster* was collected in Hyde Park, Chicago, and maintained at more than 1000 flies until testing 2-7 months later. Flies were collected from timed egg batches, recording development time. Blocks of 10 male-female pairs were mated, for mating speed, copula duration, and progeny numbers. Each fly was scored for wing length, sternopleural bristle number and all allozymes, for a total sample of 220 flies in 11 blocks. The loci measured are Adh (II-50.1), Est-6 (E6, III-36.0), Pgm (III-43.4), Est-C (EC, III-51.7), and Ao (III-56.6), giving also a measure of heterozygosity for each fly. Electrophoretic slow (S) alleles are most common at all loci. Rare alleles are combined with the less common fast (F) allele for this analysis.

A selection of the results is portrayed in Figure 1. This figure shows the average trait z-score for each genotype and sex. Additive locus effects appear as a leftward or rightward trend of the symbols. Significantly different mean scores are determined by non-overlapping error bars.

Traits show influence of loci in the following manner: **Development time.** Est-C alleles show significant additive effects, SS < FS < FF, where SS flies develop fastest. An interesting sex by genotype interaction also appears, the common tendency of male heterozygotes to develop faster than female heterozygotes, while homozygotes of both sexes tend to be similar (for Est-6, Pgm, Ao and Adh, see Figure 1). **Mate speed.** Est-6 SS males and FF females mate faster than the other genotypes. Est-C FF flies (rare, 4 of 208 flies) mate faster than the other genotypes. **Copula duration.** Adh shows an additive effect, SS > FS > FF. **Progeny.** Ao shows an additive effect, SS > FS > FF. **Wing length.** Pgm shows an additive effect, SS > FS > FF. **Bristle number.** Ao has a homozygote difference (SS > FF). Adh has a heterozygote effect (FS > FF), as does Pgm (SS > FS).

Est-6 is a known reproductive - pheromonal enzyme. Its allozymes have influenced mating speed in previous studies. The precise form of influence varies with temperature and base population characteristics. This is the first examination of Est-6 effects on mating speed within a segregating population, and it corroborates the earlier multiple population studies (Aslund & Rasmuson 1976; Gilbert 1985; Gilbert & Richmond 1982).

Loci affecting sternopleural bristle numbers have been mapped using other methods (Davies 1971). Several of these studies report loci at or very close to Pgm (within mapping resolution). At least one of these studies also report loci mapping in the Ao and Adh regions. Est-6 and Est-C fall outside of the reported bristle regions.

Based on correspondence with these previous studies, the single population method tested here is validated. This method of estimating single gene effects has many positive features that make it a very useful tool. It can be applied to any organism, including man, where mapped, polymorphic loci can be scored. It requires no artificial breeding schemes, and individuals can be sampled and scored directly from their natural population. Each individual can be scored for many traits and many loci at once. Thus, the cost and effort of this method is much less than for other methods.

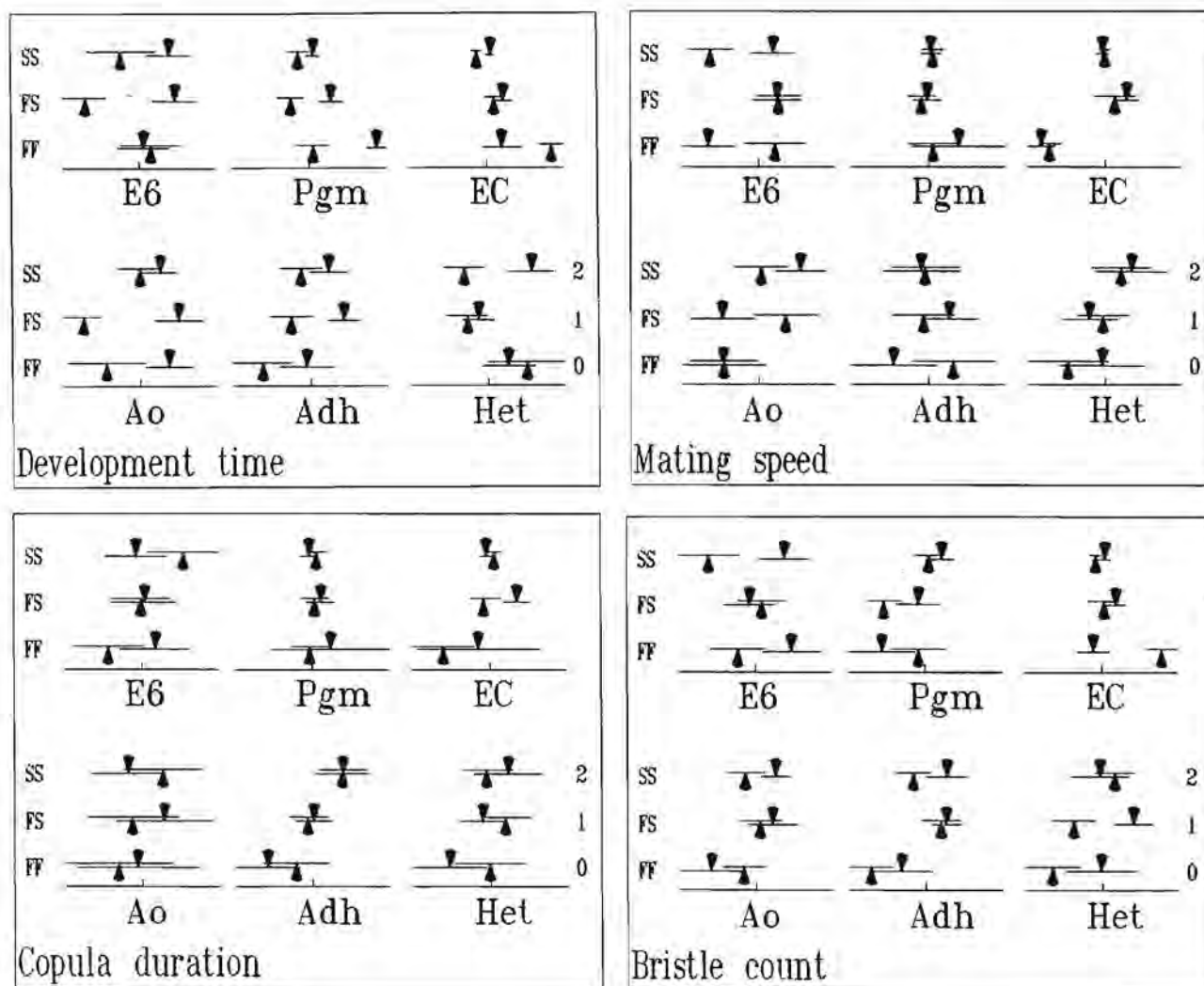


Figure 1. Average z-scores for each genotype and sex, plotted on a horizontal axis, with zero at the center. Female means are depicted with downward pointers, male means with upward pointers. The horizontal bars are the standard error of the mean. Trait means relative to the number of heterozygous loci per fly (0, 1 or 2) are shown in the subplots labeled "Het". The other subplots show scores for the three genotypes FF, FS and SS at each locus. Each locus subplot is scaled independently; the absolute dispersion is not comparable among subplots.

The use of a large, natural population minimizes error due to unknown loci linked to the measured ones. This single population method relies on the generally proven assumption that loci are in linkage equilibrium in the natural population under study. Also the method is currently limited to testing the effects of the scored loci only. A future paper will present statistical treatment and methods for applying this design to mapping quantitative trait loci at any map position.

References: Aslund, S. & M. Rasmuson 1976, *Hereditas* 62:175-178; Davies, R.W. 1971, *Genetics* 69:363-375; Gilbert, D.G. 1985, *Theor. Appl. Genet.* 69:625-629; — & R.C. Richmond 1982, *PNAS* 79:2962-2966.